

Student Observer:

Period: 

PERIOD 0 HONORS

Date:

OBSERVER INSTRUCTIONS:  
As each classmate performs their live demonstration, record the presenter(s), demo title, and physical category (Rest, Motion, or Direction). In the right column, **explain how you believe the demonstration utilizes inertia to be successful** (explain why the object maintains its state of rest or motion, how the quick action prevents unwanted force transfer, and verify that *no forward force* acts on the mass).

| DEMO & PRESENTER<br><small>Presenter Name(s), Demo Title &amp; Category</small>                                                     | HOW THE DEMONSTRATION UTILIZES INERTIA TO BE SUCCESSFUL<br><small>Explain why the mass behaves as observed using Newton's 1st Law (resistance to acceleration, <math>\Sigma F = 0</math>, zero forward force)</small> |
|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>DEMO 1<div><div></div> Rest <div></div> Motion <div></div> Direction</div><div>Presenter(s):</div><div>Demo Title:</div></div> |                                                                                                                                                                                                                       |
| <div>DEMO 2<div><div></div> Rest <div></div> Motion <div></div> Direction</div><div>Presenter(s):</div><div>Demo Title:</div></div> |                                                                                                                                                                                                                       |
| <div>DEMO 3<div><div></div> Rest <div></div> Motion <div></div> Direction</div><div>Presenter(s):</div><div>Demo Title:</div></div> |                                                                                                                                                                                                                       |
| <div>DEMO 4<div><div></div> Rest <div></div> Motion <div></div> Direction</div><div>Presenter(s):</div><div>Demo Title:</div></div> |                                                                                                                                                                                                                       |
| <div>DEMO 5<div><div></div> Rest <div></div> Motion <div></div> Direction</div><div>Presenter(s):</div><div>Demo Title:</div></div> |                                                                                                                                                                                                                       |
| <div>DEMO 6<div><div></div> Rest <div></div> Motion <div></div> Direction</div><div>Presenter(s):</div><div>Demo Title:</div></div> |                                                                                                                                                                                                                       |
| <div>DEMO 7<div><div></div> Rest <div></div> Motion <div></div> Direction</div><div>Presenter(s):</div><div>Demo Title:</div></div> |                                                                                                                                                                                                                       |
| <div>DEMO 8<div><div></div> Rest <div></div> Motion <div></div> Direction</div><div>Presenter(s):</div><div>Demo Title:</div></div> |                                                                                                                                                                                                                       |

| DEMO & PRESENTER<br>Presenter Name(s), Demo Title & Category              | HOW THE DEMONSTRATION UTILIZES INERTIA TO BE SUCCESSFUL<br>Explain why the mass behaves as observed using Newton's 1st Law (resistance to acceleration, ΣF = 0, zero forward force) |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEMO 9 [ ] Rest [ ] Motion [ ] Direction<br>Presenter(s):<br>Demo Title:  |                                                                                                                                                                                     |
| DEMO 10 [ ] Rest [ ] Motion [ ] Direction<br>Presenter(s):<br>Demo Title: |                                                                                                                                                                                     |
| DEMO 11 [ ] Rest [ ] Motion [ ] Direction<br>Presenter(s):<br>Demo Title: |                                                                                                                                                                                     |
| DEMO 12 [ ] Rest [ ] Motion [ ] Direction<br>Presenter(s):<br>Demo Title: |                                                                                                                                                                                     |
| DEMO 13 [ ] Rest [ ] Motion [ ] Direction<br>Presenter(s):<br>Demo Title: |                                                                                                                                                                                     |
| DEMO 14 [ ] Rest [ ] Motion [ ] Direction<br>Presenter(s):<br>Demo Title: |                                                                                                                                                                                     |
| DEMO 15 [ ] Rest [ ] Motion [ ] Direction<br>Presenter(s):<br>Demo Title: |                                                                                                                                                                                     |

HONORS SYNTHESIS: BRIDGING INERTIA TO MECHANICAL EQUILIBRIUM (ΣF = 0)

1. Minimizing Unwanted Force Transfer: Which demonstration observed today achieved the cleanest isolation of inertia? Explain how the presenter minimized contact time (Δt) and friction to prevent unwanted force from disturbing the mass:

2. The Equilibrium Condition: In every demonstration observed today, whenever the mass stayed at rest or moved at constant velocity, what was the horizontal net force acting on that mass? Write the equation: ΣF<sub>x</sub> = \_\_\_\_\_ N. What happens the instant an unbalanced net force acts?

Demonstrations Logged: \_\_\_\_ / 15      Honors Synthesis Check: [ ] Complete      Teacher Verification: \_\_\_\_\_